

Deep Budget-Deferral Guide

Scientific Data Analysis v2.5

Hardening the modular family while preserving public paths

Internal health: thin entrypoints, deferred implementation bodies, internal helpers, and the recorded plugin-eval A/95 result

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Translation of the preserved private v2.5 guide at commit d6583f1. Original TeX SHA-256: 08040631e64ed58e00e7a472d7f744af6c71f9add402accac15cfd1c6c684c1. Private source: `skills/scientific-data-analysis/.cache/active_references/guides/guia_scientific_data_analysis_v2_5.tex`.

Recorded results and closure steps are historical, not current verification. This release guide does not synchronize or modify Scientific Workbench; the app remains outside its scope. Historical commands are retained for traceability: use fresh output directories when reproducing them, without overwriting earlier evidence. The original remains unchanged privately.

Abstract

`scientific-data-analysis` v2.5 continues v2.4. v2.4 stabilized the mother skill; v2.5 extends that work across `scientific-data-analysis`, `scientific-data-astro`, `scientific-data-documents`, `scientific-data-notebooks`, and `scientific-data-maintainer`. It defers heavy content without deleting capabilities or undertaking a large rewrite. Public scripts remain stable entrypoints; substantial executable bodies live in `fixtures/`, and internal helpers in `fixtures/_internal/`. The mother’s substantial historical content becomes compact stubs or external maintenance archives. Its duplicate registry becomes a stub resolving the canonical maintainer registry. The intended result is that the skills remain substantial domain tools, while `evaluate-skill` rates them A/95 and `improve-skill` finds no required fixes. A documented static warning about deferred cost remains accepted. Scientific Workbench is not modified.

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1. Executive overview

v2.5 changes internal architecture, not capabilities. It aims to let the scientific-data family grow with less pressure on the context budget, without turning each child into another oversized skill.

The conservative product decisions are:

- `scientific-data-analysis` remains the normal public entrypoint.
- `scientific-data-astro` remains the expert child for FITS, WCS, photometry, spectra, RV, and optional or legacy astronomy routes.
- `scientific-data-maintainer` remains maintainer-only.
- Public scripts remain in `scripts/`.
- Substantial implementation bodies move to `fixtures/`.
- Substantial internal helpers move to `fixtures/_internal/`.
- The `public_surface_registry.yaml` public contract is preserved.
- Scientific Workbench is not modified.
- The canonical 61-row registry lives in maintainer; the mother keeps a compact stub resolving it for the gates.

The original guide both says that the registry is unchanged and describes replacing the mother’s duplicate with a stub. These statements refer to the preserved public contract and the changed storage/resolution layout, respectively.

2. From v2.4 to v2.5

Area	v2.5 change
Mother	<code>app_run_bundle.py</code> and <code>skill_hygiene_check.py</code> become thin entrypoints with bodies in fixtures.
Astro	Scripts become thin entrypoints dispatching to same-name fixtures.
Astro shims	<code>latex_workbench.py</code> and <code>notebook_workbench.py</code> remain shims to documents/notebooks, not fixture wrappers.
Documents	Scripts become thin entrypoints with executable bodies in <code>fixtures/</code> .
Notebooks	Scripts become thin entrypoints with executable bodies in <code>fixtures/</code> .
Maintainer	Scripts become thin entrypoints with bodies in <code>fixtures/</code> .
Internal helpers	<code>provenance_utils</code> , <code>run_bundle</code> , <code>runtime_common</code> , <code>tabular_io</code> , and related helpers become compact shims.
Ownership matrix	The full v2.3 matrix is archived in <code>fixtures/archived_references/</code> ; the active reference is a compact stub.
Maintainer history	Substantial historical matrices and inventories are archived; active references are compact stubs.
Mother router	The full v2.3 module map is archived outside the active skill; the router uses compact deterministic metadata when that external file is absent.
Maintainer module map	Its v2.3 copy also becomes a canonical stub to stay below the plugin-eval static threshold.

Mother history	Substantial historical contracts and plans live outside the active skill; active paths retain compatibility stubs.
Historical guides	Non-current PDF and TeX guides are archived outside the active skill; v2.5 becomes active.
Mother registry	<code>public_surface_registry.yaml</code> becomes a compact stub resolving the canonical maintainer registry.
Historical documents	v1.7/v1.9/v2.0/v2.2/v2.3/v2.4 references become navigation stubs where full content is already archived.
Mother wrappers	Avoid unsafe imports of private helpers from children.
Plugin-eval	All five skills reach A/95 without mandatory required fixes.
Documentation	Adds <code>v2-5-budget-deferral-hardening.md</code> and this v2.5 guide.

3. What remains unchanged

- No new capability or public path removal.
- No changes to scientific contracts, artifact types, errors, statuses, or app hints.
- No forced general-purpose analogy for astronomy or legacy routes.
- No edits to original user data.
- No synchronization or development of Scientific Workbench.

4. v2.5 architecture

The pattern is:

```
scripts/inspect_fits.py # stable entrypoint
fixtures/inspect_fits.py # substantial executable body
scripts/_internal/fixture_script_dispatch.py
scripts/_internal/provenance_utils.py # compact shim
fixtures/_internal/provenance_utils.py # substantial internal body
scripts/_internal/deferred_internal.py # internal loader
```

Users and gates continue to call `scripts/*.py`. Implementation changes belong in the corresponding fixture. This reduces routine reading cost while preserving historical compatibility.

4.1. Shim rule

Some scripts re-export a tool from another child and have no local implementation. In astro:

- `latex_workbench.py` continues to delegate to `scientific-data-documents`.
- `notebook_workbench.py` continues to delegate to `scientific-data-notebooks`.

Converting those shims into fixture wrappers would be incorrect. v2.5 retains them as shims and verifies that behavior through regression checks.

5. Plugin-eval evidence

The important v2.5 result is closure of mandatory work and visible code recommendations across all five skills. The final recorded installed measurement, after compacting mother and maintainer with margin, was:

Installed skill	Score	Grade	Deferred
scientific-data-analysis	95	A	44913
scientific-data-astro	95	A	49754
scientific-data-documents	95	A	28155
scientific-data-notebooks	95	A	28040
scientific-data-maintainer	95	A	46230

All five retained the *deferred cost* warning.

All five `deferred_cost_tokens` measurements remain in the heavy band. The original release accepts this because actual domain implementations, wrappers, contracts, and maintenance evidence remain available. Mandatory fixes and warnings about complexity, long lines, and long functions were closed. Remaining backlog:

- The static `deferred_cost_tokens-budget-high` warning.
- Formal coverage, if coverage metrics are desired later.

6. v2.5 regressions

Main regression:

```
python scripts/audit_v2_5_budget_deferral_regression.py
```

It validates:

- Representative astro wrappers and fixtures.
- Astro shims that must remain shims.
- Representative documents, notebooks, and maintainer wrappers.
- Internal helpers deferred to `fixtures/_internal/`.
- Mother maintenance wrappers.
- The active historical matrix as a compact stub and its complete canonical fixture archive.
- Compact historical maintainer references with archived canonical bodies.
- A compact mother router module map that remains loadable at runtime.
- The compact maintainer module-map copy and its archived canonical body.
- Compact mother historical contracts/plans and their archived canonical bodies.
- The mother's duplicate registry replaced by a canonical stub to maintainer.
- Compact, synchronized README, RELEASE, and public snapshot documents.
- Mother wrappers preserving their own private helpers.

Earlier gates remain in place:

```
python scripts/audit_v2_4_budget_architecture_regression.py
python scripts/audit_v2_3_release_candidate_regression.py
python scripts/audit_v2_3_module_ownership_regression.py
```

7. Historical closure criteria

The original v2.5 closure required:

1. `guia_scientific_data_analysis_v2_5.tex` compiles to PDF.
2. The PDF and TeX are copied to the Desktop for that historical local handoff.
3. `audit_v2_5_budget_deferral_regression.py` passes.
4. Affected v2.3/v2.4 regressions pass.
5. Documents/notebooks wrapper and internal-helper smoke checks pass.
6. `sync_public_surface_docs.py --check` passes.
7. The editable skill is synchronized to the installed family only after green gates.
8. Post-synchronization gates pass.
9. `evaluate-skill` and `improve-skill` run on installed mother and children.

8. Recommended usage after v2.5

Ordinary usage remains unchanged:

```
python scripts/scientific_workflow_router.py plan input --task "..."  
python scripts/profile_table.py data.csv --summary-json summary.json  
python scripts/inspect_fits.py image.fits --summary-json summary.json
```

For maintenance, release work, and plugin-eval:

```
python scripts/audit_v2_5_budget_deferral_regression.py  
python scripts/audit_v2_4_budget_architecture_regression.py  
python scripts/sync_public_surface_docs.py --check
```

9. Conclusion

v2.5 is an internal improvement rather than a visible user feature. The modular architecture now defers substantial execution bodies, internal helpers, and historical material as well as separating responsibilities. It is intended to support growth while preserving compatibility and reducing pressure on usage budgets. The original guide's aspiration about weekly usage limits is not a measured account-level saving; its evidence is the static plugin-eval result documented above.